

Your Talk Title:
Short Descriptive Phrase

Venue or Challenge Name — Team / Group

Author One Author Two Author Three

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Mohammed VI
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**College of
Computing**

Problem Overview

Setting

Task: one-line description of the dataset / benchmark

Core difficulty: the failure mode you care about

- ▶ Failure mode A
- ▶ Failure mode B
- ▶ What a good solution must do

Approach

Strategy: high-level idea in one sentence

Base model: MODEL with your features / backbone

Change: what you add on top

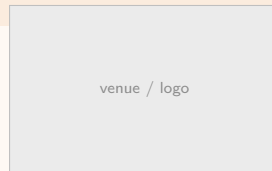


Figure 1: Venue

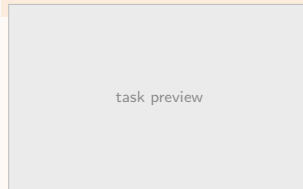


Figure 2: Task preview

Family A

Variant 1:

- ▶ Setup / pretraining
- ▶ Segmenter / head
- ▶ Outcome: metric and failure mode

Variant 2: improved but still **below target**

Family B

- ▶ What it models well
- ▶ Where it breaks (F1 / edit / boundaries / ...)

Motivation for Model

One sentence: why this architecture sits between the two families.

Notes

- ▶ Takeaway from family A
- ▶ Takeaway from family B
- ▶ What you still need

High-level approach

- ▶ Inputs / features
- ▶ Temporal or spatial model: MODEL
- ▶ How the branches / modules exchange information
- ▶ How you decode at inference

How it works

- ▶ **Module A:** local evidence
- ▶ **Module B:** long-range / global context
- ▶ **Coupling:** attention, fusion, or gating
- ▶ **Output:** framewise / pixelwise / scalar prediction



Figure 3: MODEL overview

Loss formulation

Term 1: $\mathcal{L}_1 = \frac{1}{N} \sum_i \ell(y_i, \hat{y}_i)$

Term 2: $\mathcal{L}_2 = \frac{1}{N} \sum_i w_i r_i$

Combined: $\mathcal{L} = \mathcal{L}_1 + \alpha \mathcal{L}_2 + \beta \mathcal{L}_3$

Intuition

- ▶ What each term encourages
- ▶ How weights / gates behave
- ▶ Inference cost (usually: none)



Figure 4: Where the extra head / loss sits

Dataset & Features

Split: train / val / test, number of classes

- ▶ Resolution, fps, or input size
- ▶ Feature / backbone dimensionality
- ▶ Sampling stride or augmentation

Training

Initialization: pretrained / warm start

- ▶ Learning rate
- ▶ Key scalar (merge weight, temperature, ...)

Search: W&B / grid / random

- ▶ Hyperparameter A: $\{a_1, a_2\}$
- ▶ Hyperparameter B: $\{b_1, b_2\}$

Hardware

- ▶ GPU model
- ▶ Wall-clock / memory notes

Search spaces

- ▶ β : {1.0, 1.5, 2.0}
- ▶ γ : {2.0, 3.0}
- ▶ α : {1.0, 2.5, 5.0}
- ▶ Other knobs held fixed

Selected configuration

$\beta = \dots, \gamma = \dots, \alpha = \dots$



Test set

Submission / hold-out:

- ▶ **Headline metric:** xx% (rank / vs. baseline)
- ▶ Consistency note
- ▶ What the proposed variant changes

Validation

Method	Acc	F1	Edit
Baseline	00.0	00.0	00.0
+ proposed	00.0	00.0	00.0

Qualitative

- ▶ Cleaner transitions / fewer artifacts
- ▶ Remaining errors

Key insights

- ▶ What actually moved the metric
- ▶ Cost at train vs. test time
- ▶ What must be true for the gain to show up

sequence 1: GT vs. prediction

Figure 5: Ground truth (top) vs. predictions (bottom)

Component analysis

Each add-on:

- ▶ Term A alone
- ▶ Term B alone
- ▶ Combined: best (+x metric)

Training dynamics:

- ▶ Initialization / warm start
- ▶ Learning-rate sensitivity

Sensitivity

- ▶ Best γ , β , merge weight, ...

Cost

Training: extra head / loss only

Inference: identical to baseline

Limitations

- ▶ Single dataset
- ▶ Feature / backbone choice
- ▶ Supervision type

Contributions

Method:

- ▶ Architectural change
- ▶ Loss / training change
- ▶ Train-only overhead

Results:

- ▶ Headline number
- ▶ Qualitative gain

Impact

- ▶ Who uses this
- ▶ Why the cost profile matters

Future work

- ▶ Alternative integrations
- ▶ Other datasets
- ▶ Deployment

One-sentence takeaway that can be photographed from the back row.

Thank You

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